PH142: Introduction to Probability and Statistics in Biology and Public Health

Fall 2026

Course format: Live online lectures, in-person labs and exams

4 semester credits

Please note: This syllabus is subject to change. The course website will contain the most current schedule, links, and assignment information: https://ph142-ucb.github.io/fa26/.

Course Format and Meeting Information

Course	PBHLTH 142
Class number	31216
Instruction mode	Live online lectures, in-person labs and exams
Lecture	Monday, Wednesday, and Friday, 8:00–9:00 AM Pacific, Online. Recordings will be posted after the lectures.
Discussion platform	Ed Discussion
Labs	In-person

Section	Class #	Days	Time	Place
101B	31254	Thursday	5:00–6:59 PM	247 Dwinelle
102B	31255	Wednesday	2:00–3:59 PM	2030 Valley Life Sciences
103B	31256	Friday	9:00–10:59 AM	2030 Valley Life Sciences
104B	31257	Wednesday	3:00–4:59 PM	251 Dwinelle
105B	31258	Wednesday	4:00–5:59 PM	250 Dwinelle
106B	31259	Wednesday	5:00–6:59 PM	246 Dwinelle
107B	31260	Thursday	4:00–5:59 PM	246 Dwinelle
108B	31261	Wednesday	1:00–2:59 PM	2062 Valley Life Sciences
109B	31262	Wednesday	9:00–10:59 AM	213 Undergrad Academic Bldg
112B	31362	Thursday	12:00–1:59 PM	2032 Valley Life Sciences

Course Description

This course is an introduction to statistics and data science, primarily for MPH and undergraduate public health majors, and others interested in public health topics.

The course can be divided into three parts.

Part I focuses on using R to explore datasets and summarize univariate and bivariate distributions visually and numerically. We will also introduce concepts around sampling and how study data are acquired. Specifically, we will use the `dplyr` and `ggplot2` packages to manipulate and visualize data sets in R.

Part II introduces classical problems in probability and the Normal, Binomial, and Poisson distributions. We also introduce concepts around sampling variability and the Central Limit Theorem.

Part III introduces statistical inference, the process of using statistics calculated from samples to make inference about populations.

During all parts of the course we will use real and simulated data sets to gain experience conducting biostatistical analyses using R. We will follow the PPDAC model, which stands for "Problem, Plan, Data, Analysis, and Conclusion".

Prerequisites

No specific coursework is required, and no programming background is necessary. We expect a level of math equivalent to a US secondary school algebra course.

Course Learning Objectives

After successfully completing Part I of the course, you will be able to:

- Extract relevant statistical information from published articles in the scientific and popular press
- Describe distributions of variables visually and calculate summary statistics for centrality and spread
- Determine appropriate graphics to plot distributions and provide R code to manipulate and visualize data frames
- Identify basic sampling strategies and study designs used in public health
- Describe core concepts of ethics in public health
- Perform basic data manipulation in R

After successfully completing Part II of the course, you will be able to:

- Compute probabilities using general probability rules
- Understand statistical independence
- Identify and describe Binomial and Poisson random variables
- Compute probabilities using basic properties of the Normal distribution
- Express epidemiologic measures as probabilities
- Describe sampling variability and the Central Limit Theorem
- Write R code to compute probabilities for the Normal, Binomial, and Poisson distributions

After successfully completing Part III of the course, you will be able to:

- Estimate means, proportions, and differences between means and proportions; compute their confidence intervals and perform statistical tests
- Choose appropriate measures and statistical tests for associations between variables

- State the assumptions and importance of the assumptions for statistical tests
- Describe and check the assumptions for simple linear regression
- Interpret confidence intervals and statistical tests for regression intercept and slope coefficients
- Write R code snippets to generate confidence intervals, perform hypothesis tests, and calculate p-values
- Formulate a clear statistical question, address it with an appropriate method, and accurately and clearly report findings

Instructor Information and Communication

Instructor: Xiudi Li (he/him/his), email: xiudil@berkeley.edu
Instructor availability: Please book through email.
Course email for non-content inquiries: 142gsi@berkeley.edu

Graduate Student Instructors (GSIs)

While the instructor will interact with the whole class and will oversee all activities and grading, the GSIs will be your main point of contact. Your GSIs are responsible for assisting you directly with questions about assignments and course requirements. They will lead lab sections and will host office hours. They will also facilitate ongoing discussion and interaction with you on major topics in each module. Communication with the teaching team regarding course content should be directed towards the Ed Discussion board.

Office Hours

GSI office hours will be posted on the course calendar. You may attend the office hours of any GSI regardless of your lab section. Instructor office hours will primarily be offered by appointment through email.

Course Mail

All questions about course material should be asked on Ed Discussion. This allows us to reduce email and also allows other students to benefit from the questions and answers. We will not answer course-content questions by email. For questions and concerns *not* related to course content, email 142gsi@berkeley.edu. Do *not* use the bCourses emailing system for course communication.

Students with Disabilities

If you are eligible for disability-related accommodations through UC Berkeley's Disabled Students' Program, please send us your letter of accommodation through the DSP student portal as soon as possible, so we can ensure your accommodations are met.

If you have any questions about your accommodations for PH142, please email 142gsi@berkeley.edu. Do not directly email any medical documentation to the PH142 teaching team. If your accommodation status changes at any point during the semester, please send us an updated letter of accommodation through the DSP student portal.

If you believe you may be eligible for disability-related accommodations, please visit the Disabled Students' Program website to begin an application as soon as possible. The application process may take several weeks.

Course Materials and Technical Requirements

Course website: https://ph142-ucb.github.io/fa26/. The course website will contain the syllabus, assignment descriptions, course resources, the most up-to-date schedule, and assignment information.

We will use R, RStudio, and Datahub. Use of R, RStudio, and Datahub is required for homework assignments and lab exercises and requires an internet connection and web browser. You will learn how to use R, RStudio, and Datahub during the first week of classes.

Optional textbook: *The Practice of Statistics in the Life Sciences* by Brigitte Baldi and David S. Moore. The 4th edition is the latest one, but previous editions are fine. The textbook is not required; course materials take precedence where there are differences.

Learning Activities and Assignments

All times listed are Pacific Time.

Lectures

Synchronous video lectures will be held on Mondays, Wednesdays, and Fridays from 8:10 to 9:00 am. Lectures will be recorded and made available as soon as possible following each session. Although lecture attendance is not recorded, attendance is strongly recommended. Please keep in mind that there may be some delay between live lecture and the posting of the lecture videos. Lectures will primarily cover material in the lecture slides and handouts. Code used during lecture will be shared in markdown format through the datahub, slides will be shared in PDF format prior to lecture. Lectures will introduce concepts, highlight examples of these concepts in the popular and scientific press, and walk through some examples and calculations. Students will be muted and their videos turned off by default since there are so many of you! I encourage you to ask questions using the chat or by unmuting and asking a question aloud. Feel free to turn on your camera when you ask a question orally. Please note that questions asked orally become part of the lecture/lab recording.

Lab/Discussion Sections

Lab/discussion sections will be held each week. These sections will be held in person for 2 hours each. Lab sessions are not recorded. Lab sections will go through hands-on coding activities in R, review key concepts and include group activities and support for group data projects.

Quizzes

Weekly quizzes will be available for 24 hours (Thursday 11:59pm - Friday 11:59pm). Quizzes will be relatively short and meant to encourage you to keep on top of weekly content. Once opened, you will have 1 hour to complete the quiz. Quizzes will cover material through the Wednesday of the current week. Your lowest quiz score will be dropped.

If you have a Letter of Accommodation at UC Berkeley, confirm with your GSI that it has been received and accommodations have been made. Every time you start a quiz or exam, check to confirm you have the correct time accommodation. If not, notify your GSI.

Lab Assignments

Lab assignments are intended to practice concepts from lecture in a practical programming environment. They are intended to be completed during the lab section with support and troubleshooting help from the GSIs. If you are unable to attend lab on a given week you may complete the lab independently. Lab assignments are generally released on Mondays prior to the first lab section, and are due by Saturday at noon. Lab assignments are submitted through Datahub to Gradescope and are graded on correct completion. This means the grading is all or nothing. Assignments which are fully complete and pass all tests for code correctness will be counted 100%. Incomplete or incorrect assignments are counted as a 0%. Assignments that are complete and correct which are submitted after the due date but within 24 hours of the due date will be accepted for 50%. There are 11 lab assignments total. You may miss one lab assignment without penalty.

Midterms I and II and final exam

There are two midterms and one final exam. If you have a conflict with any of the exam dates, please email the instructor before September 1st so that we can discuss possible accommodations. Appropriate accommodations for the midterm will be made for those with disabilities (please refer to the "Disabilities" section). Please contact the instructor as soon as possible if your accommodation for extended time may create a conflict with another scheduled class. Please note that only in extremely rare circumstances such as illness (with a doctor's note) will the midterm be given to individual students after the scheduled examination date. Exams will cover the material presented in lecture, supplemental videos, discussion, and lab sections, including R coding syntax, unless otherwise noted.

Midterms and final policies. The midterm and final will allow a "cheat sheet" of notes. You may bring one page of handwritten or printed notes with you to the midterms and two pages of handwritten or printed notes with you to the final. If your notes are printed the font should be a minimum of 10 point font. Your notes page will be turned in with the exam. If you want to have your notes to use later you should make a separate copy to keep that you do not bring to the exam. You may not use the internet to search for the answers or inform your answers. Using the internet or AI tools is strictly prohibited and any evidence of this may result in a zero on the exam. While you take the exam, you are prohibited from discussing the test with anyone other than the PH142 instructional team. Evidence of cheating may result in a zero on the test or further disciplinary action. Exams will be given in person, on paper and scanned into gradescope by the teaching team. We will strive to return graded examinations within one week of the exam date.

The midterms will take place on October 2 and 30. The final will take place on December 14.

Data Skills Demonstration Group Project

The purpose of the group project is to use public health or biological data that you find or have access to and use it to demonstrate statistical concepts learned throughout the course. Data project groups are randomly assigned by the teaching team and will be released on Friday, September 4. Students are required to check in with their lab GSI prior to submitting each part of the data project for participation credit.

The project will be submitted in three parts. The Fall 2026 deadlines are:

- Part I: September 25 at 11:59 PM Pacific.
- Part II: October 23 at 11:59 PM Pacific.
- Part III: December 4 at 11:59 PM Pacific.

Participation

Participation is worth 10% of your overall course grade. Your participation grade consists of lab attendance (7%) and three data project GSI check-ins (3%). You will earn full credit for the lab attendance portion by attending at least 9 of the 11 lab sections.

Optional Homework Assignments

Homework problem sets are optional and not submitted for credit. These assignments provide practice and preparation for exams, labs, and the data project. Solutions will be posted after students have had time to complete the assignment.

Grading

The final course grade percentages are:

Category	Percentage	Notes
Quizzes	10%	9 quizzes total, lowest grade dropped
Lab Assignments	10%	11 lab assignments total, and you may miss one without penalty
Midterm exam I	15%	
Midterm exam II	15%	
Final exam	20%	
Data project	20%	Submitted in 3 parts
Participation	10%	Includes lab section attendance (7%) and data project check-ins with your GSI (3%)

Course Schedule

Please refer to the course website https://ph142-ucb.github.io/fa26/ for the most up-to-date course schedule.

Date	Lectu #	Topic	Lab (W/Th/F)	Reading	Assignments and dates
Week 1					
8/24/2026	–	No lecture	Lab 01: Introduction to R and RStudio on DataHub	–	–
8/26/2026	1	Introduction to the course, the cloud, and PPDAC	–	–	Lab 01 released; HW 01 released
8/28/2026	2	Working with data in R and RStudio (`dplyr` package)	–	–	Lab 01 due Saturday
Week 2					
8/31/2026	3	Visualizing data in R and RStudio (`ggplot2` package)	Lab 02: Visualization of global Cesarean delivery rates	–	Lab 02 released; HW 02 released
9/2/2026	4	Visualizing distributions for one variable; numerically summarizing spread and central tendency	–	Chs. 1–2	–
9/4/2026	5	Exploring relationships between two variables	–	Ch. 3	Lab 02 due Saturday; Quiz 01 due Friday
Week 3					
9/7/2026	–	No lecture: holiday	Lab 03: Relationship between global Cesarean rates and GDP	–	Lab 03 released; HW 03 released
9/9/2026	6	Introduction to regression	–	Ch. 4	–
9/11/2026	7	Two-way tables (relationships between two categorical variables)	–	Ch. 5	Lab 03 due Saturday; Quiz 02 due Friday
Week 4					
9/14/2026	8	Samples and observational studies	Lab 04: Problem set on probability calculations	Ch. 6	Lab 04 released; HW 04 released
9/16/2026	9	Live exercise: Sampling births from U.S. territories	–	–	–
9/18/2026	10	Designing experiments	–	Ch. 7	Lab 04 due Saturday; Quiz 03 due Friday
Week 5					
9/21/2026	11	Introduction to probability	Midterm I review session	Ch. 9	Midterm I review problems released (optional)
9/23/2026	12	General rules of probability	–	Ch. 10	–
9/25/2026	13	General rules of probability, continued	–	Ch. 10	Data Project Part I due
Week 6					
9/28/2026	14	The Normal distribution, Part I	Lab 05: Sensitivity, specificity, and the Normal distribution	Ch. 11	Lab 05 released; HW 05 released
9/30/2026	15	The Normal distribution, Part II	–	Ch. 11	–
10/2/2026	–	Midterm 1	–	–	Lab 05 due Sunday; Quiz 04 due Saturday
Week 7					
10/5/2026	16	The Binomial distribution	Lab 06: Problem set on Normal, binomial, and Poisson distributions	Ch. 12	Lab 06 released; HW 06 released
10/7/2026	17	The Poisson distribution	–	Ch. 12	–
10/9/2026	18	Sampling distributions for a mean and proportion, and the Central Limit Theorem	–	Ch. 13	Lab 06 due Saturday; Quiz 05 due Friday

Continued on next page

Date	Lectu #	Topic	Lab (W/Th/F)	Reading	Assignments and dates
Week 8					
10/12/2026	19	Confidence intervals for a mean with known standard deviation	Lab 07: Classroom simulation on the Central Limit Theorem and confidence intervals	Ch. 14	Lab 07 released
10/14/2026	20	Hypothesis tests for a mean with known standard deviation	–	Ch. 15	–
10/16/2026	21	Power, Type I and Type II error, and sample size (Part I)	–	Ch. 15	Lab 07 due Saturday; Quiz 06 due Friday
Week 9					
10/19/2026	22	Inference for a population mean with unknown standard deviation	Midterm II review session	Ch. 17	Midterm II review problems released (optional)
10/21/2026	23	Catch up if behind	–	Ch. 17	–
10/23/2026	24	Comparing two means	–	Ch. 18	Data Project Part II due
Week 10					
10/26/2026	25	Matched comparisons	Lab 08: Paired and two-sample t-tests	Ch. 17	Lab 08 released; HW 07 released
10/28/2026	26	Inference for a population proportion	–	Ch. 19	–
10/30/2026	–	Midterm 2	–	–	Lab 08 due Sunday; Quiz 07 due Saturday
Week 11					
11/2/2026	27	Comparing two proportions	Lab 09: Inference for proportions	Ch. 20	Lab 09 released; HW 08 released
11/4/2026	28	Bootstrapping confidence intervals	–	–	–
11/6/2026	29	The chi-square test for goodness of fit	–	Ch. 21	Lab 09 due Saturday; Quiz 08 due Friday
Week 12					
11/9/2026	30	The chi-square test for two-way tables	Lab 10: Problem set on the chi-square test	Ch. 22	Lab 10 released; HW 09 released
11/11/2026	–	No lecture: holiday	–	–	–
11/13/2026	31	Permutation tests	–	–	Lab 10 due Saturday
Week 13					
11/16/2026	32	Inference for regression I	Lab 11: Regression models and checking model assumptions	–	Lab 11 released; HW 10 released
11/18/2026	33	Inference for regression II	–	Ch. 23	–
11/20/2026	34	Comparison of many means (ANOVA)	–	Ch. 24	Lab 11 due Saturday; Quiz 09 due Friday
Week 14					
11/23/2026	35	ANOVA II/Tukey's HSD	Holiday; no lab	–	–
11/25/2026	–	No lecture: holiday	–	–	–
11/27/2026	–	No lecture: holiday	–	–	–
Week 15					
11/30/2026	36	Nonparametric testing alternatives	Final exam review session	–	Final review problems released (optional)
12/2/2026	37	Regression modeling with a categorical exposure	–	–	–
12/4/2026	38	Final exam review	–	–	Data Project Part III due
Week 16: RRR Week					
Week 17: Finals Week					
12/14/2026	–	Final exam	–	–	–

Due Dates, Late Work, and Regrades

Due dates for all graded assignments will be listed on the course website and communicated in announcements on Ed Discussion.

Late work. Assignments submitted within 24 hours after the due date will be accepted with a 50% penalty. Extensions can be made for DSP students and should be requested by emailing the course email. Other extension requests should explain the situation and must include documentation.

Regrades. Regrade requests must be submitted through Gradescope within three days after grades are released. Regrades are for correcting an error in the application of the published rubric. If you request reconsideration of one part of an assignment, instructors may reconsider grades on the entire assignment. Due to the short turnaround time for final grade submissions, we generally cannot accommodate regrade requests for the final exam.

Academic Integrity

You are a member of an academic community at one of the world's leading research universities. Any test, paper, code, or report submitted by you and bearing your name is presumed to be your own original work unless otherwise specified. You may use words or ideas from other individuals, publications, websites, or other sources only with proper attribution.

Students may not circulate or post course materials, including handouts, exams, syllabi, assignments, or solutions, without written permission from the instructor. If you are unclear about expectations for completing an assignment or taking an exam, seek clarification from the instructor or a GSI beforehand.

Strategies for Successful Learning

Based on past experience with this course, we recommend that you:

- Attend lab consistently
- Ask questions in lecture, lab, office hours, and Ed Discussion
- Study in a group and take turns explaining concepts to each other
- Review questions and concepts from weekly assignments and exams where you lost points
- Use practice exams to become familiar with question types and to practice the material

Take Care of Yourself

Do your best to maintain a healthy lifestyle during this semester by eating well, exercising, getting enough sleep, and taking time to recharge your mental health. If you start to feel overwhelmed, be kind to yourself and reach out for support. Seeking help is a courageous thing to do for yourself and for those who care about you.

Support resources include emotional, physical, safety, social, and basic wellbeing resources for students. Academic resources can be found at the Student Learning Center and English Language Resource sites. Berkeley's Office of Emergency Management has resources to prepare for emergencies.

We also ask that you look out for your fellow peers. If you see any signs that may indicate your classmate may need assistance, please use the resources above or reach out to any of the GSIs or the instructor.

Incomplete Course Grade and Course Evaluation

Students who have substantially completed the course but, for serious extenuating circumstances, are unable to complete remaining course activities may request an Incomplete grade. This request must be submitted in writing to the GSI and instructor and must include verifiable documentation.

Before the course ends, please participate in the course evaluation. Your feedback helps improve the course and future instruction.